ISOM5240

Group Project

Timelines:

Project submission (deadline): (refer to the syllabus)

Project presentation: (refer to the syllabus)

Grading Criteria:

- Project Objectives (Deep Learning Oriented)
- Data Pre-processing
- Design and Implementation of Business Application
- Logical Approach
- Use of Python
- Report and Presentation

Submission:

1. **Project Report (Page limit: < 10): (in pdf file)**

   Your project report must include, at a minimum, the following sections.

   - **Project title and student names**: This section includes the official project title and the names of the students responsible for its completion.
   - **Company Name and Website URL**: Your project is tailored for a particular company. Please specify the company's name and its website URL.
   - **Project Objective (not exceeding 50 words):** Outline the business problem(s) targeted for resolution in this project. The focus is on leveraging deep learning techniques.
     (For example: This project aims to conduct sentiment analysis for improved understanding of customer feedback, facilitating data-driven decision-making in the organization.)
   - **Strategy**: Outline the approach to effectively achieving the stated project objective.
   - **Model URL:** Fine-tuned model on Hugging Face
   - **App URL:** Deployed app (**Public**) on Streamlit Cloud
   - **GitHub URL:** GitHub Repository linked to Streamlit Cloud app
   - **Dataset**: Describe your dataset in point form and highlight selected features.
     (a) Number of labels:
     (b) Relevant features:
     (c) Number of samples:
       - **Training** (for finetuning):
       - *Validation (if applicable)*:
       - **Testing**:
     (d) Sources:
       e.g., [Provide URLs or sources for the dataset, such as https://huggingface.co/....]
   - **Model**:
     Utilize Hugging Face's pre-trained models along with their fine-tuned counterparts for the project, specifying their names, e.g., **deepseek-ai/DeepSeek-R1**.
     Illustrate and elucidate the structure of your model with diagrams that yield optimal results.
       e.g., **image-to-text** -> **text generation** -> **text-to-audio**

- **Deployment:**
  Incorporate screenshots along with descriptions to illustrate the usage of your application in Streamlit Cloud.
- **Experiments**:
  (a) Experiments encompass two main objectives: (i) selecting the optimal model(s), and (ii) assessing the overall performance of the application on Streamlit Cloud.
    a. Model selection: This involves evaluating accuracy and runtime exclusively.
    b. Application performance on Streamlit Cloud: Accuracy is the sole metric considered in this evaluation.
  (b) Models: Outline the models that were explored and describe the process leading to the optimal (or final) model or model pipeline.
  (c) Tabulate the experimental results, giving special attention to using a reasonable sample size during testing to ensure robust and reliable outcomes.

- **Conclusion**: Summarize your findings and elucidate
  (a) whether you have successfully addressed the stated business problem(s) or not
  (b) how to implement your application for resolving the stated business problems.

2. Python notebooks (in a folder)

(i) For fine-tuning: Colab Notebook programs (in .ipynb file)

Colab Notebook(s): Present your program logic step by step in your Colab notebooks with comments. **Keep the program outputs in the notebooks.** When grading your project, we will click "**Restart & Run All**," and your program should execute and produce outputs/results identical to those in your report. **(Note: Programs failing to run will result in a lower mark. It is highly recommended that students test their programs before submission.)**

Fine-tune models to precisely align with the project's objectives. Each student is required to define one primary project objective and fine-tune only one model accordingly. If students opt for multiple project objectives, it is likely that the project will necessitate fine-tuning across several pre-trained models using multiple highly relevant datasets.

(ii) For testing and experiments: Use Colab notebook programs saved in .ipynb file

3. App files (in a folder)

All application files, including app.py, requirements.txt, etc., are required.

4. Dataset files (in a folder)

Ensure the incorporation of all necessary data files, including image folders, essential for your program. This encompasses testing data, data for fine-tuning, and data for assessing model performance.

5. Fine-tuned Model files (in a folder)

All fine-tuned models utilized in the app must be submitted.

6. Presentation File and Video (in PPT and mp4)

- The project presentation is scheduled for the final day of this course and should not exceed 10 minutes.
- Groups must deliver their project presentations using a PowerPoint file, which should incorporate images and diagrams.
- Submission requires a video presentation file in mp4 format, where students' faces need to be visible.
- If your group is chosen for a presentation, you will present your project in person.

7. Experimental Results (in Excel file)

Record all experiments, conducted with a reasonable number of testing samples, and present the results in an Excel file.

Submission Guidelines

1. Compress all your files into a zip archive and name it using your group number and the group leader's student ID, for example, Group01_07123456.zip.
2. Student groups must name their video files using their two-digit group number (e.g., grp01.mp4) and submit them to a designated cloud location before the deadline.
3. Submit the zip file through Canvas. In the case of an excessively large file, please notify our TA via email, providing a downloadable link (e.g., URL) for file retrieval.
4. Multiple submissions are permitted; however, only the last submission will be evaluated. Ensure that your final submission is not only the best but also functional.
5. Late submissions within a 3-hour window will incur a penalty of a 30% deduction from your total marks. No further late submissions will be accepted.
6. Plagiarism (Copying) is a grave concern. Students may explore the Internet for information, references, or programs, but a stringent rule is in place: COPYING MORE THAN SEVEN WORDS from a source is considered cheating. The MINIMUM penalty for this offense is zero marks for the project.

Submission Checklist

1. Project report
2. Python notebooks
3. GitHub app files
4. Dataset files
5. Fine-tuned model files
6. Presentation files
7. Experimental results
8. Streamlit Cloud App URL

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├── Canvas/
│   ├── Group01_documentation/
│   │   ├── Project_report.pdf
│   │   └── Experimental_results.xlsx
│   ├── Group01_program/
│   │   ├── Python_notebooks/
│   │   │   ├── Model1.ipynb
│   │   │   ├── Model2.ipynb
│   │   │   ├── Model3.ipynb
│   │   │   └── ...
│   │   ├── GitHub_App_Files/
│   │   │   ├── app.py
│   │   │   ├── requirements.txt
│   │   │   └── ...
│   ├── Group01_Dataset_files/
│   │   ├── data.csv
│   │   └── Fine-tuned_Model_files/
│   │   └── ...
│   └── Group01_presentation/
│   │   ├── Presentation_slide.pptx
│   │   └── grp01.mp4
│   └── Streamlit Cloud App URL (Website URL)
└── OneDrive/
    └── Group01_video/
        └── grp01.mp4
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Grading Rubric

	Excellent	Average	Poor
Business Application			
Business Objective	Objectives are exceptionally clear, measurable, and highly practical.	Objectives are somewhat clear, measurable, and practical.	Objectives are unclear, not measurable, or impractical.
Practicality	Addresses significant business challenges with innovative deep learning applications.	Addresses some business challenges, with moderately innovative applications.	Fails to address relevant business challenges or lacks innovation.
Applicability	Developed a fully functional, user-friendly, and impactful business application.	Developed a functional but less user-friendly or impactful application.	Developed an application that is either non-functional or not user-friendly.
Model			
Efficiency	Selected model(s) are highly efficient, optimizing resources while maintaining performance.	Selected model(s) are moderately efficient, with some resource optimization.	Selected model(s) are inefficient, with poor resource optimization.
Effectiveness	Model(s) demonstrate exceptional accuracy and effectiveness in solving the business problem.	Model(s) show adequate accuracy and effectiveness with room for improvement.	Model(s) lack accuracy or effectiveness, failing to adequately solve the business problem.
Suitability	The chosen model(s) are highly suitable for the specific business problem, demonstrating a deep understanding of the problem space.	The chosen model(s) are somewhat suitable but may not fully address the nuances of the problem.	The chosen model(s) are unsuitable for the problem, showing a lack of understanding.
Expected Results			
Achieved Objective	Objectives are fully met with outcomes exceeding user expectations.	Objectives are partially met, with outcomes meeting user expectations.	Objectives are not met, or outcomes fall short of user expectations.
Result	Achieves outstanding results demonstrating the potential of the solution.	Achieves satisfactory results with room for improvement.	Results are inadequate, showing limited potential for the solution.
Coding			
Python Utilization	Python is utilized effectively, showcasing deep understanding and creativity.	Python is utilized adequately, though lacks advanced features or creativity.	Python usage is basic or incorrect, demonstrating limited understanding.
Logical Problem-Solving	Design and implementation processes are logical, efficient, and innovative.	Processes are logical but may lack efficiency or innovation.	Processes are illogical, inefficient, or lack coherence.
Error-Free Operation	The code operates flawlessly, with thorough testing and error handling.	The code operates with minor errors or issues that do not significantly impact functionality.	The code contains significant errors that hinder functionality.
Documentation and Presentation			
Documentation	Comprehensive documentation adhering to guidelines, thoroughly explaining all aspects.	Adequate documentation but lacks detail or adherence to guidelines.	Incomplete or poorly structured documentation, failing to meet guidelines.
Presentation	Presentation is clear, engaging, and effectively communicates all key points.	Presentation is clear but lacks engagement or misses some key points.	Presentation is unclear, unengaging, or fails to communicate key points.

Important Notes:

- No bonus points will be awarded for any additional features or functions, as the grading rubric above will be uniformly applied to evaluate all projects.
- Any discrepancies in files will incur penalties. For instance, the model utilized in Streamlit Cloud app differs from the one generated by your program for fine-tuning.
- Students using models not covered in the lectures may need to address their technical difficulties themselves.
- Due to the tight schedule, the deadline will NOT be extended.
- A project must comprise **at least two pipelines**. A "pipeline" is defined as a Hugging Face pipeline that utilizes a pre-trained model.
- Pre-trained models must be those provided on the Hugging Face platform. Other pre-trained models, such as those from OpenAI, cannot be used in the projects.
- Projects involving large language models (LLMs), such as DeepSeek, are **NOT** recommended due to the limited availability of GPU resources and technical support. Additionally, these projects often require substantial time and effort for development and fine-tuning. All projects will be assessed using the same grading criteria. Therefore, student groups are strongly advised to consult with the professor before pursuing such projects.